

Task 3 – Network Packet Analysis Using Wireshark

Objective

The objective of this task is to capture and analyze network packets using Wireshark in order to identify protocols, inspect packet structures, and detect unusual or suspicious traffic.

Tools Used

- Wireshark
 - Kali Linux
 - Web Browser
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Introduction

Network packet analysis helps cybersecurity professionals monitor and inspect communication between devices in a network. By analyzing packets, it is possible to identify protocols, connection behavior, and potential security issues.

In this task, Wireshark was used to capture HTTP, DNS, and TCP traffic and analyze packet structures and flags.

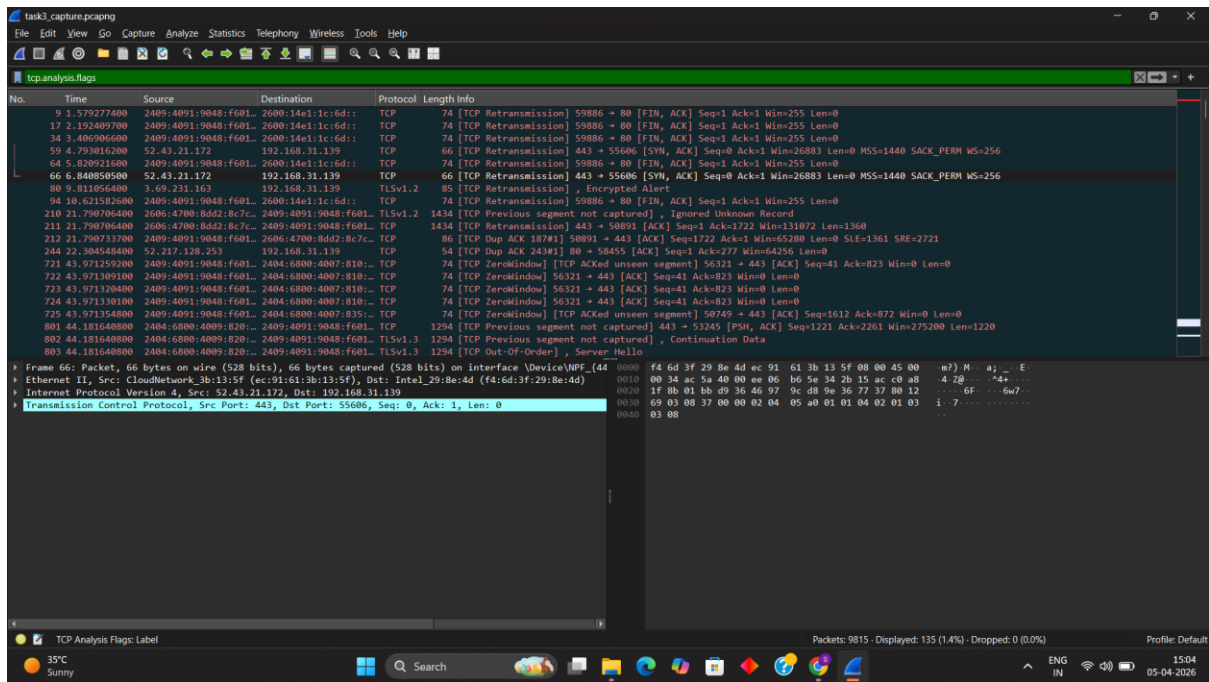
Step 1 – Capturing Network Traffic

Procedure

- Opened Wireshark
- Selected the active network interface
- Started packet capture
- Generated traffic by browsing websites and accessing online services

Findings

Wireshark successfully captured live network packets exchanged between the local system and external servers.



Step 2 – DNS Traffic Analysis

Objective

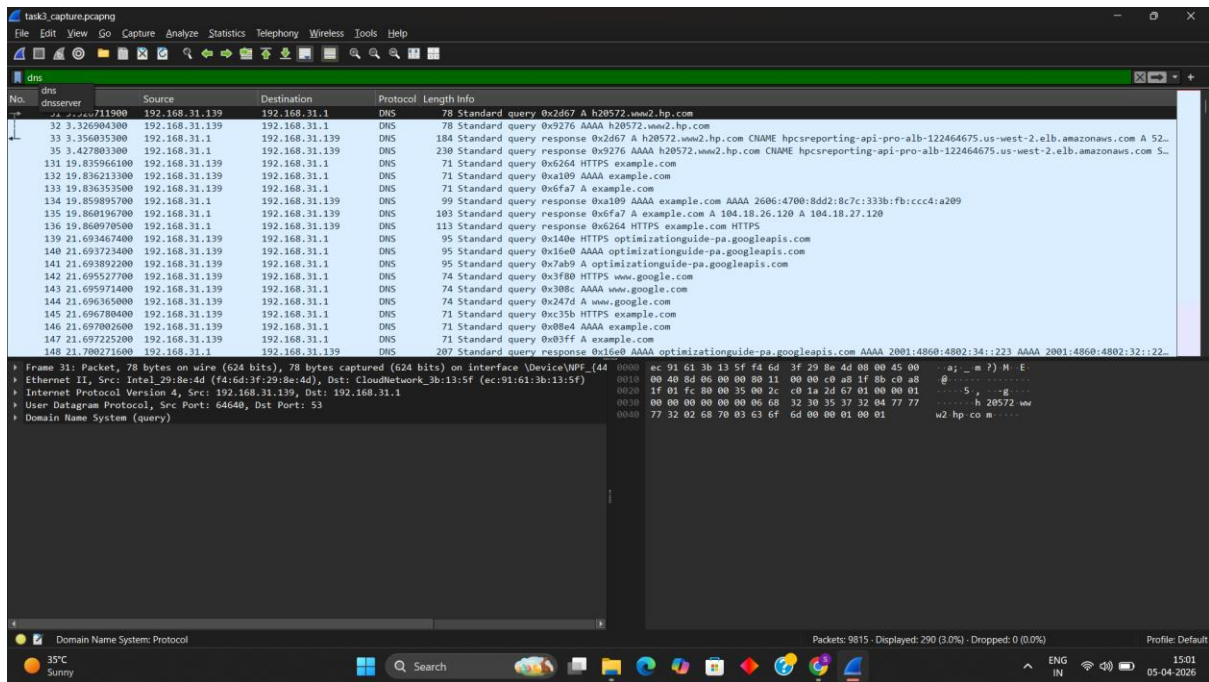
Analyze DNS request and response packets.

Findings

- DNS packets were identified during domain name resolution.
- Source and destination IP addresses were visible.
- Domain queries were successfully captured.

Analysis

DNS traffic helps systems convert domain names into IP addresses. Unencrypted DNS traffic may expose browsing activity.



Step 3 – TCP Handshake Analysis

Objective

Analyze TCP connection establishment.

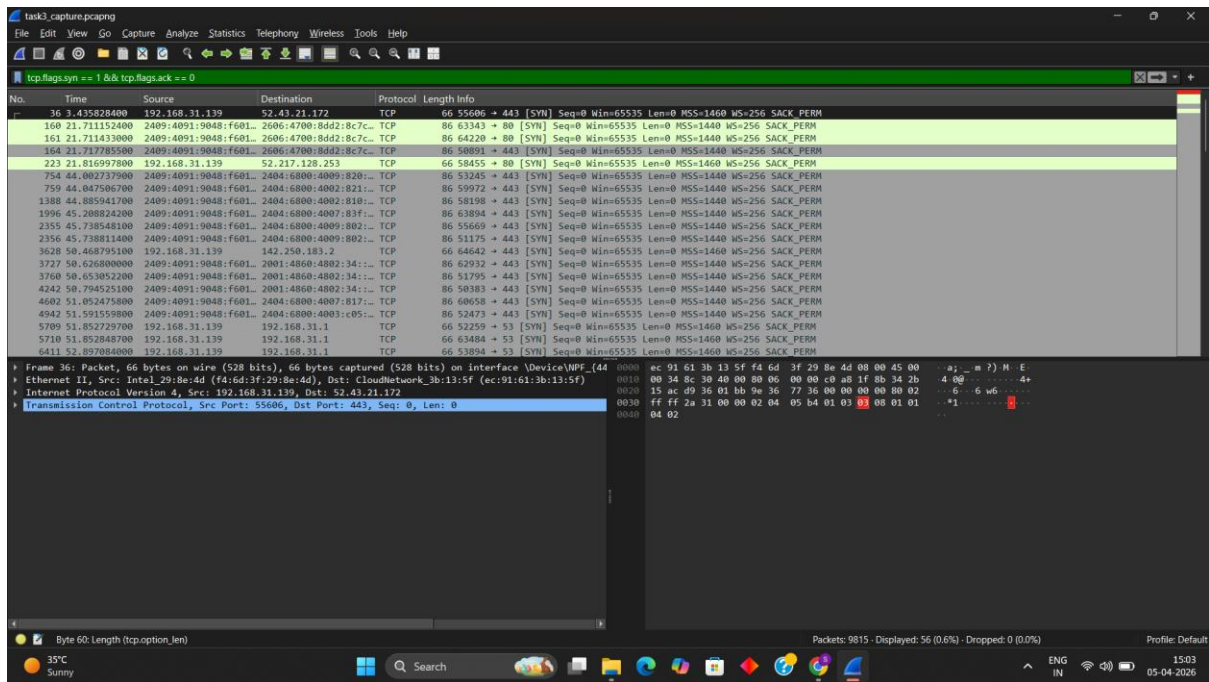
Findings

The TCP three-way handshake was observed:

1. SYN packet sent by client
2. SYN-ACK packet returned by server
3. ACK packet sent by client

Analysis

TCP handshakes establish reliable communication between systems before data transfer begins.



Step 4 – HTTP Packet Analysis

Objective

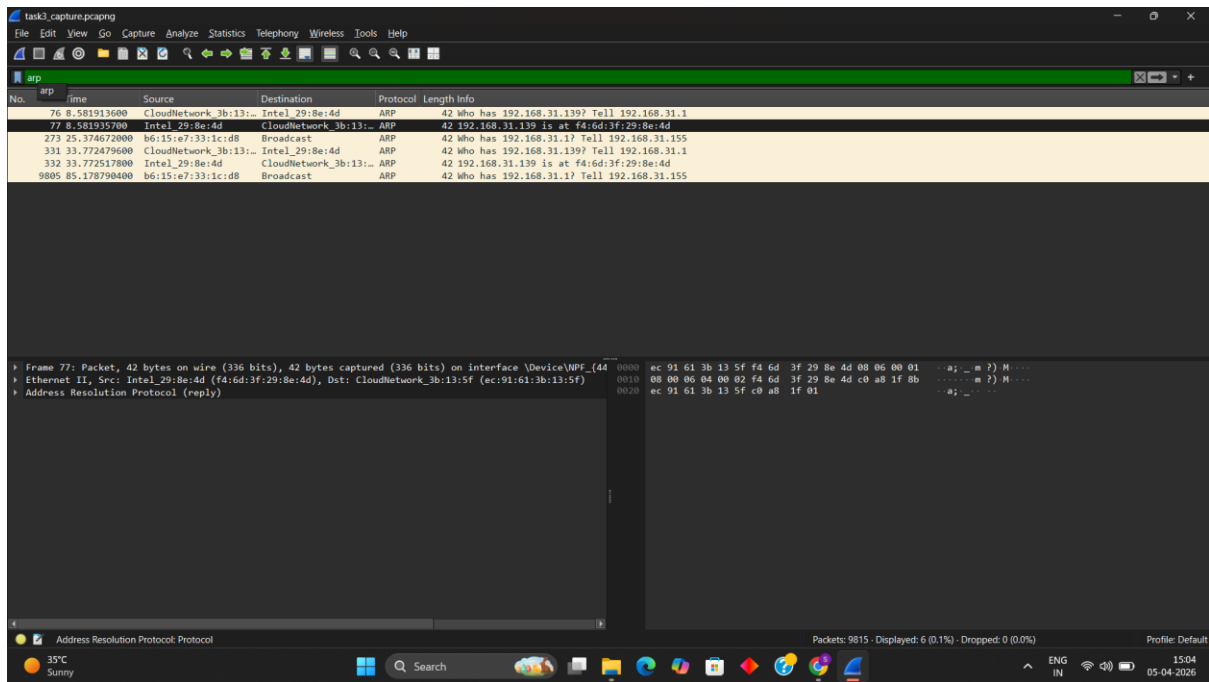
Inspect HTTP traffic and packet structure.

Findings

- HTTP GET requests were captured.
- Request headers and response information were visible.
- Communication occurred over port 80.

Analysis

HTTP traffic is unencrypted and may expose sensitive information during transmission.



Step 5 – Suspicious or Unusual Traffic Identification

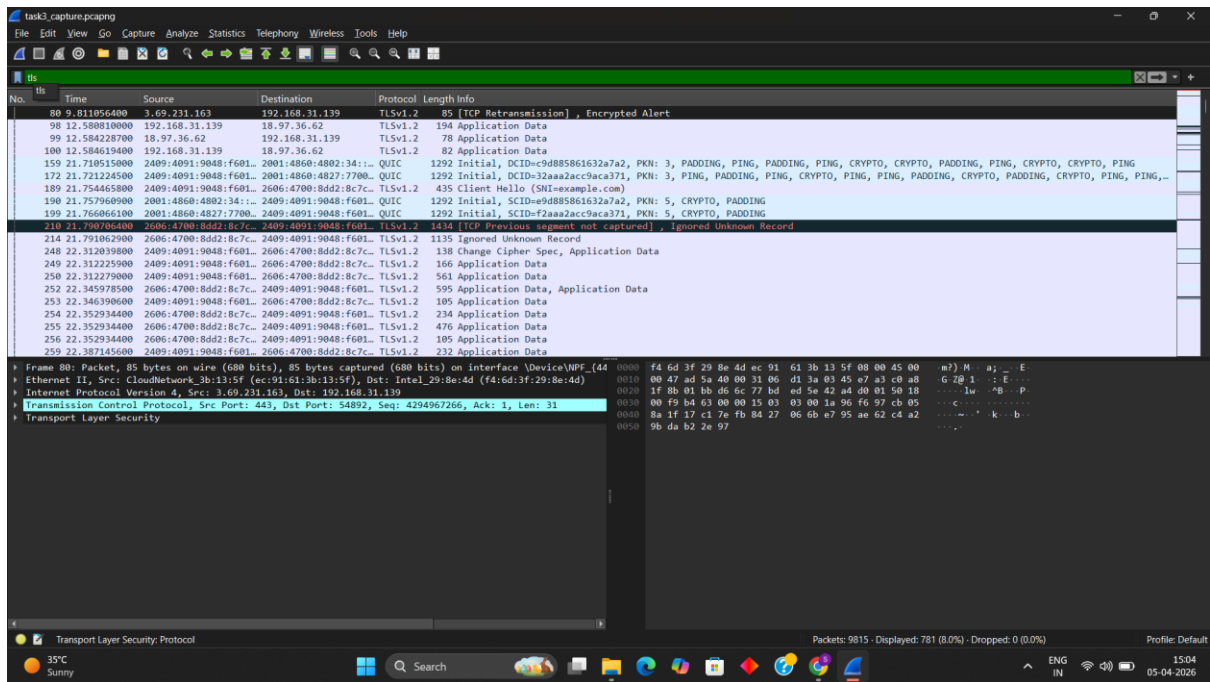
Findings

- Some packets showed retransmissions and duplicate acknowledgments.
- Unusual packet flags may indicate network congestion or connection issues.

Analysis

Malformed or suspicious packets can sometimes indicate:

- Network instability
- Misconfigurations
- Potential malicious activity



Packet Structure and Flags Observed

Protocol Important Fields Observed

DNS	Query name, transaction ID
TCP	SYN, ACK, sequence numbers
HTTP	GET request, headers
ARP	Source MAC, destination MAC

Mitigation Recommendations

Issue Recommendation

Unencrypted HTTP Traffic	Use HTTPS/TLS encryption
DNS Privacy Exposure	Use secure DNS services
Suspicious Packets	Monitor network traffic regularly
Network Congestion	Troubleshoot network performance

Conclusion

This task demonstrated how Wireshark can be used to capture and analyze network traffic. Packet analysis helps identify protocols, monitor communication behavior, and detect suspicious or unusual network activity.

Ethical Statement

All packet captures and analyses were performed in a safe and authorized environment strictly for educational purposes only.